

Literature Status Note: Non-density of Minimal Weak Phase Retrieval Frames

Source arXiv:2110.06868; answer arXiv:2301.03520

17 August 2026

Status: literature already answered. Problem 9.1 of arXiv:2110.06868 is explicitly answered by Theorem 6 of arXiv:2301.03520. This is a provenance record, not a new solution claim.

Source question

F. Akrami, P. G. Casazza, A. Rahimi, M. A. Hasankhani Fard, and B. Daraby, *A note on (weak) phase and norm retrievable Real Hilbert space frames and projections*, arXiv:2110.06868 (2021), Section 9, Problem 9.1, page 14:

Show that the weak phase retrievable frames with $2n - 2$ vectors in $\mathbb{R}^n$ are not dense in all frames with $2n - 2$ vectors.

Supporting answer

P. G. Casazza and F. Akrami, *Classifying weak phase retrieval*, arXiv:2301.03520v2 (2023), Section 5, Theorem 6, page 7, proves:

The family of m -element weak phase retrieval frames is not dense in the set of m -element frames in $\mathbb{R}^n$ for every $m \geq 2n - 2$.

Taking $m = 2n - 2$ gives exactly Problem 9.1. The supporting paper's abstract also explicitly says that this result answers a longstanding open problem, so the relation is author-recognized rather than an agent-only implication.

Scope

This literature theorem answers only the non-density question. It does not settle Problem 9.2 (existence of $2n - 2$ vector examples for $n \geq 5$) or Problem 9.3 (weak phase retrieval by $2n - 2$ or $2n - 3$ proper subspaces). A separate candidate partial packet records an independent cofactor characterization, an algebraic-nullity strengthening in the minimal case, and a correction to the cited $\mathbb{R}^4$ example.

Evidence and search bound

The source PDF and supporting PDF are stored with this note. The identification was confirmed from the exact theorem statement and the supporting abstract. A bounded search of the parsed arXiv corpus and current arXiv/web results found no later source needed to establish the status.

References

- [1] F. Akrami, P. G. Casazza, A. Rahimi, M. A. Hasankhani Fard, and B. Daraby, *A note on (weak) phase and norm retrievable Real Hilbert space frames and projections*, arXiv:2110.06868 (2021).
- [2] P. G. Casazza and F. Akrami, *Classifying weak phase retrieval*, arXiv:2301.03520 (2023).